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7 November 1985
SUPERSEDING
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MILITARY SPECIFICATION

# CHILLED WATER UNITS, BAKERY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers mechanical water cooling units for bakeries.

1.2 Classification. Water cooling units shall be of the following types and sizes (see 6.2).

Type I - Ingredient water cooler.

Size 1 - 20 gallons per hour, 1 horsepower.
Size 2 - 30 gallons per hour, 1-1½ horsepower.

Type II - Combination ingredient water and jacket water coolers.

Size 1 - 30 gallons per hour, 3 horsepower.
Size 2 - 70 gallons per hour, 5 horsepower.
Size 3 - 100 gallons per hour, 7½ horsepower.
Size 4 - 120 gallons per hour, 10 horsepower.

## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants

TT-P-664 - Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting

PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted

MIL-P-116 - Preservation, Methods of

MIL-C-3774 - Crates, Wood; Open, 12,000 and 16,000-Pound Capacity

MIL-S-6872 - Soldering Process

MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignition

MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment

STANDARDS

FEDERAL

FED-STD-H28 - Screw Threads Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-279 - Compressors, Reciprocating Power-Driven, Open-Type, For Use with Refrigerant-12

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of

MIL-STD-759 - Condensers, Air Cooled, Refrigerant-12

MIL-STD-773 - Compressors, Hermetic: For Use with Refrigerant-12

MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods

MS 17291 - Receivers, Liquid, Refrigerant-12

MS 17415 - Valve, Safety Relief, Refrigerant-12

MS 17797 - Valves, Regulating, Fluid Pressure (Water), Refrigerant-12

MS 17798 - Indicator, Sight, Liquid Refrigerant-12

MS 17854 - Valves, Regulation, Fluid Pressure, Refrigerant-12



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MS 17982 - Valves, Expansion, Thermostatic Refrigerant-12 and Refrigerant-22
MS 27211 - Thermometer, Indicating, Capillary Tube and Bulb, 2½ Inch Dial/Size
MS 27900 - Gages, Pressure and Gages Compound, Pressure and Vacuum; Dial Indicating, Panel Mounted, 2½ Inch Dial Size, Refrigerant R12 and R22
MS 35845 - Dehydrators, Desiccant, Refrigerant-12
MS 35867 - Connection End, Internal Flare, Refrigeration
MS 35868 - Connection End, External Pipe Thread; Refrigeration
MS 35869 - Adapter, Straight male Refrigeration Pipe to Flared Tube
MS 35870 - Adapter, Refrigerant Cylinder, External Flare Straight Pipe, Refrigeration
MS 35871 - Tees, Pipe To Tube, Refrigeration
MS 35872 - Flare Nut and Flare Seal Bonnet, Refrigeration
MS 35873 - Connection End, External Flare, Refrigeration
MS 35880 - Valves, Packless Stop, Solder-Joint, Connection Ends
MS 35881 - Valves, Packless, Stop, Flare Connection Ends
MS 35882 - Valves, Packless, Stop, Three-Way, Solder-Joint Connection Ends
MS 35883 - Valves, Packless, Stop, Three-Way, Flare Connection Ends
MS 35917 - Elbows, Tube, 90 and 45 Degree, Solder Joint, Refrigeration
MS 35918 - Adapter, Straight, Pipe Thread to Tube, Male Solder-Joint Refrigeration
MS 35919 - Adapter, Straight, Pipe to Tube, External Flare to Female Solder-Joint, Refrigeration
MS 35920 - Reducer Coupling, Tube, Male to Female Solder-Joint, Refrigeration
MS 35921 - Adapter, Straight, Pipe to Tube, Pipe Thd to Female Solder-Joint, Refrigeration
MS 35922 - Elbow, Tube, 90 Degree, Reducing, Solder-Joint, Refrigeration
MS 35923 - Reducer, Coupling Tube, Female to Female Solder-Joint Refrigeration
MS 35924 - Plug and Cap, Tube, Solder-Joint, Refrigeration
MS 35825 - Connection End, Solder-Joint, Refrigeration
MS 35926 - Connection End, Internal Pipe Thread, Refrigeration
MS 35927 - Coupling, Tube, With Stop, Solder-Joint Refrigeration
MS 35928 - Elbows, Tube, Street, 90 and 45 Degree, Solder-Joint Refrigeration
MS 35929 - Tees, Tube, Straight, Solder-Joint, Refrigeration
MS 49008 - Tees, Tube, Flare Connections, Refrigeration



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DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

5-11-649 - Chilled Water Units, Bakery; Schematic.

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS, INC. (ASHRAE)

Standard No. 15 - Safety Code for Mechanical Refrigeration
Standard No. 30 - Methods of Testing for Rating Liquid Chilling Packages

(Application for copies should be addressed to the American Society of Heating, Refrigerating, and Air Conditioning Engineers, Inc., 345 East 47th Street, New York, N.Y. 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standard MG 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, N.Y. 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1621 - Compressive Properties of Rigid Cellular Plastics
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.



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3. REQUIREMENTS

3.1 Standard product. Chilled water units delivered shall be the manufacturer's current standard models except for any changes necessary to comply with the requirements of this document. All accessories and components normally furnished commercially as shown by current catalogs and sales brochures, shall be furnished with the chilled water units offered under this document and shall be of the same material, finish, quality, and quantity as furnished commercially with the same standard product.

3.2 Standard compliance. Chilled water units shall conform to the applicable requirements of ASHRAE Standard No. 15 for safety (see 4.3).

* 3.3 Design. The chilled water units shall consist of all the components shown on Drawing 5-11-649 and as specified herein, and any additional components required to meet the requirements specified in 3.3.1. The type I chilled water unit shall have one ingredient water cooler. The type II chilled water unit shall consist of an ingredient water cooler and a jacket water cooler. The chilled water units, shall each be completely assembled as a packaged unit and shall employ the dichlorodifluoromethane refrigerant as specified in 3.4.3.

3.3.1 Operating characteristics. Sizes and capacities for type I and type II chilled water units shall be in accordance with table I and table II, when tested as specified in 4.4.1.

* 3.4 Materials (see 6.3) and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial chilled water units provided the completed item complies with all the provisions of this document.

* 3.4.1 Valves and fittings. Valves and fittings for the refrigeration system shall conform to MS 35867 through MS 35873, MS 35880 through MS 35883 and MS 35917 through MS 49008, as applicable.

3.4.2 Fastening devices. Bolts, nuts, screws, washers, and cotter pins used in the construction and mounting of components, in the sizes and type recommended by the contractor, shall conform to current commercial standards. Threads shall conform to FED-STD-H28.

* 3.4.3 Refrigerant. The refrigerant shall be dichlorodifluoromethane conforming to type 12 of BB-F-1421.

* 3.4.4 Insulation. Insulation shall be non moisture absorbing material; shall be homogeneous throughout; and shall have "K" factors as specified below using a 75°F mean temperature and expressed in terms of British thermal units (Btu) per hour per square foot per inch of thickness per degree F temperature difference (W/M.K.). Loose fill type insulation will not be acceptable. the insulation



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TABLE I. Type I chilled water units
Ingredient water cooler

| Size | Gal./hr. (min.) | Storage gal. (min.) | Water-in temp. (min.) | Water-out temp. (max.) | Cond. unit Btu/hr. (nominal) | Unit hp (nominal) |
|---|---|---|---|---|---|---|
| 1 | 20 | 30 | 80°F | 35°F | 10,000 | 1 |
| 2 | 30 | 45 | 80°F | 35°F | 15,000 | 1-½ |

TABLE II. Type II chilled water unit
Combination ingredient water and jacket water coolers

| | Ingredient Water | | | | Jacket Water | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Size | Gal./hr. (min.) | Water in temp. (min.) | Water out temp. (max.) | Storage gal. (min.) | Storage gal. (min.) | Water in temp. (min.) | Water out temp. (max.) | Circulation gal. (min.) | Cond. unit Btu/hr. | Unit HP (nominal) |
| 1 | 30 | 80°F | 35°F | 45 | 30 | 38°F | 35°F | 4 | 20,000 | 3 |
| 2 | 70 | 80°F | 35°F | 100 | 40 | 38°F | 35°F | 6 | 40,000 | 5 |
| 3 | 100 | 80°F | 35°F | 150 | 50 | 38°F | 35°F | 10 | 60,000 | 7-½ |
| 4 | 120 | 80°F | 35°F | 200 | 60 | 38°F | 35°F | 12 | 80,000 | 10 |





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shall be of a type that is not combustible or will not support combustion and will not release toxic fumes when exposed to high temperature.

* 3.4.4.1 Semirigid. Semirigid insulation shall be "B" fiber or finer industrial type fiberglass with a manufacturer's commercial rated density of at least 1.0 but not more than 3.0 pounds per cubic foot and shall be held together with a thermosetting binder.

* 3.4.4.2 Pre-expanded. Pre-expanded insulation shall be polystyrene or polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.26 for polystyrene, and 0.15 for polyurethane with a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.4.3.

* 3.4.4.3 Foamed-in-place. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.15, and a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.4.3.

3.5 Design and Construction.

3.5.1 Chilled water units. Each chilled water unit shall be completely assembled as a packaged unit ready for operation, after connection to utilities. Electrical characteristics shall be as specified (see 6.2).

3.5.2 Welding. Surfaces to be welded shall be cleaned free of rust, scale, paint, grease and other foreign matter. Welds shall not be scaled, blistered, have flux residue, or be oxidized. All weld beads, fillets, and butts shall be of sufficient size to develop strength equal to that of the members joined. Metal projections and rough edges shall be ground to eliminate bumps and uneven or sharp edges.

3.5.3 Housing. The steel frame for the housing shall be constructed of angles not less than 1-1/2 by 1-1/2 by 1/8-inch. The water cooler section shall be covered with sheet steel panels not less than 20 gage (0.0359 inch) thick. A section of the housing shall be removable to provide easy access to the water coolers. The machine section shall be provided with removable grill type panels to provide sufficient ventilation and allow dissipation of heat.

3.5.4 Insulation installation. The insulation specified in 3.4.4 shall be securely installed so that it will remain in place. Vapor barrier material, except when foamed-in-place insulation is used, shall be applied to completely prevent penetration of water vapor into the insulation. Joints between insulation shall fit snugly so that heat losses will be reduced to a minimum. Metal or other heat conducting materials extending through the insulation shall be held to the absolute minimum (see 4.2.2).

* 3.5.5 Coolers. Coolers shall consist of a storage tank and evaporator. Tanks and evaporators shall consist of any one of the following combination:



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Separate evaporate (cooler) and storage tank, closed pressure tank with submerged evaporator, or enclosed surface type coil with tank. The cooler tank shall be fabricated from stainless steel. The tank shall not leak (see 4.4.1). If, of the closed pressure type, the tank shall not leak when hydrostatically tested at 300 pounds per square inch gage (p.s.i.g.) (see 4.2.2). The water flow-control valves for ingredient water coolers shall be electrically operated and wired for remote control. The flow-control valves for the jacket coolers shall be of the manual type. A drain connection shall be provided in each tank, for complete gravity drainage. Each cooler shall be insulated and housed in a tight galvanized steel container with removable cover affording ready access to cooling tank.

3.5.6 Evaporators. Evaporator coils shall be pipe or tube fabricated of stainless steel or copper. Evaporator coils may be finned. Metals of the evaporator and tank if dissimilar, shall be protected to resist electrolytic or galvanic corrosion. Each evaporator shall be provide with an adjustable thermal expansion valve in accordance with MS 17982 and a thermostatically operated solenoid liquid shut off valve with provision for manual opening.

3.5.6.1 Dehydrator. A dehydrator conforming to MS 35845 shall be installed in the liquid line with a valved by-pass including receiver outlet valve, to make it possible to by-pass the dehydrator to facilitate replacement.

3.5.6.2 Liquid indicator. A liquid and moisture indicator conforming to MS 17798, shall be installed in the liquid line in a readily visible and accessible position.

3.5.6.3 Refrigerant solenoid valve. A refrigerant solenoid valve thermostatically operated, shall be provided in the liquid line to each evaporator. The solenoid valve shall have provisions for manual opening or a three-valve by-pass of the solenoid valve shall be installed.

3.5.6.4 Back-pressure controls. An adjustable, back-pressure regulating valve conforming to MS 17854 shall be provide in the refrigerant suction line from each evaporator. A pressure gage and a three-valve by-pass around each back pressure regulating valve shall be provided.

3.5.7 Condensing unit. The condensing unit shall be air cooled, water cooled, or evaporative type as specified (see 6.2). Each unit shall be complete and shall include compressor, condenser, receiver, motor, motor starter, safety devices, and all accessories necessary for operation. Each condensing unit shall be mounted on vibration dampening pads and over a water tight condensation drip pan constructed of galvanized sheet metal, with a 2-inch lip, and provided with a drain connection. The refrigeration system after being charged with refrigerant and oil shall not contain moisture in excess of 25 parts per million (p.p.m.) when tested as specified in 4.4.2. There shall be no leakage when tested as specified in 4.2.5.2. Condensing unit shall be sized for maximum of 2 gallons of condenser water per minute, per ton of refrigeration based on water



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inlet temperature of 80°F and refrigeration condensing temperature of 105°F and a 26°F suction temperature.

3.5.7.1 Compressor. The compressor shall be open type, hermetic, or semi-hermetic. Open type compressors shall conform to MIL-STD-279, and hermetic and semi-hermetic compressors shall conform to MIL-STD-773. The operating speed of the compressor shall not exceed the compressor manufacturer's established speed for the intended service. Open type compressors shall be equipped with an oil filler plug and positive shaft-sealing device which shall be replaceable in its entirety without the necessity of refinishing the crank shaft shoulder.

3.5.7.2 Compressor motor. The compressor motor shall have a capacity to start the refrigerating system under full load. Motors shall be protected against damage from overloads and voltage fluctuations when starting or running.

3.5.7.2.1 Motor starter. Automatic starters shall be supplied with 3-phase compressor motors. The compressor motor starter shall be across-the-line type, continuous duty, suitable for automatically starting the condensing unit compressor motor when actuated by the automatic temperature control device. The motor starter shall be equipped with a thermal overload relay protector if the motor is not otherwise equipped with such protection. All overload relays shall be calibrated for the particular motor application.

3.5.7.3 Pressure gages. A refrigerant suction pressure gage and a refrigerant discharge pressure gage conforming to MS 27900 shall be provided with each condensing unit. Gages shall be connected to the back seating connections of compressor service valves.

3.5.8 Safety devices.

* 3.5.8.1 Pressure control. A dual high and low pressure control as normally provided by the manufacturer shall be furnished.

3.5.8.2 Safety relief valve. The receiver and the condenser shall be provided with a safety relief valve conforming to MS 17415.

* 3.5.9 Condensers.

3.5.9.1 Water-Cooled. The condenser shall be supplied as a condenser receiver or as a condenser alone. The condensers shall be supplied with a drain connection to completely drain the condenser of water, when necessary, for servicing or prevention of freezing. The end plates shall be removable for cleaning. A purging connection shall be located on either the condenser or condenser-receiver at a location to remove noncondensible gasses.

3.5.9.2 Air-cooled. Air-cooled condensers shall conform to MIL-STD-759.



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3.5.10 Receiver. When a condenser alone is furnished, the receiver shall be of welded steel construction as specified in MS 17291, and shall have sufficient capacity to hold the maximum operating charge of refrigerant plus an additional 25 percent when the refrigerant is at 90°F (see 4.4.5).

* 3.5.11 Refrigerant lines. Refrigerant lines shall be copper tube. Two manually operated shutoff valves shall be provided, one at the inlet and one at the outlet connections of each receiver. Shutoff valves shall be of the packless type. Each valve shall be identified by having its functional name on a waterproof metal or plastic tag permanently attached to it. Soldering shall conform to MIL-S-6872. Line vibration or expansion loops shall be provided as required.

3.5.12 Water-piping. The water lines connecting the condenser, pumps, and water coolers shall be copper or standard weight galvanized iron pipe. The inlet to the condenser shall be provided with a regulating valve conforming to MS 17797. The waste line from the condenser shall be installed and sized as to assure full, free flow of condenser cooling water. When a jacket water cooler of the closed pressure type is furnished, the water make up line to the jacket water cooler shall be provided with an adjustable pressure regulating valve set to relieve the line pressure when it reaches 15 pounds per square inch (p.s.i.). The drain from each water cooler shall be provided with a manually operated shutoff valve and shall be extended below the supporting framework. All piping and fittings containing chilled water shall be insulated to prevent formation of condensation. The insulation shall consist of a nonsweat type pipe covering of single or double layer construction. Insulation on piping, piping supports, and fittings shall be inherently vapor resistant or treated to provide a vapor seal. The insulation shall be neatly fitted and secured (see 4.2.2).

3.5.12.1 Water filters. The type I, size 1 and size 2, and the type II, size 1 chilled water units shall be equipped with a filter rated at not less than 6 gallons per minute. Type II, sizes 2, 3, and 4 chilled water units shall be equipped with filters rated at not less than 12 gallons per minute. Location of the water filters shall be such as to provide easy access for servicing.

3.5.13 Water cooler pumps. Each water cooler shall be provided with a pump mounted on vibration dampening pads and provided with a rust-resistant condensation drip pan and a drain cock. Each pump shall be of the centrifugal type, bronze-fitted, and driven by a direct connected electric motor. Each ingredient water pump motor shall be wired so that the motor operates when the condensing unit is operating and also can be operated independently of the condensing unit. For type II chilled water units, the pumps for both coolers shall be the same make, model, and horsepower, with interchangeable parts. Pump capacities for type I and II chilled water units shall be not less than 10 gallons per minute for sizes 1 and 2 and 20 gallons per minute for sizes 3 and 4 against a 40-foot head (see 4.2.2). Pipe connections between pumps and coolers shall be not less than 1 inch iron pipe size.



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* 3.5.14 Water pump motors. Water pump motors shall conform to the NEMA Standard MG-1. Water pump motors shall be rated for continuous duty with an allowable temperature rise of 122°F under the operating conditions specified herein. Water pump motors shall start and operate the pumps satisfactorily under operating load conditions and shall be capable of starting the equipment when the terminal voltage is 90 percent of the rated voltage. Motors shall have speeds not in excess of 1,800 revolutions per minute (r.p.m.). Water pump motor controls shall be totally inclosed types. A starter for each pump motor shall be provide and equipped with thermal overload protection and manual reset device.

3.5.15 Wiring. All wiring within the chilled water unit enclosures shall be installed in rigid or flexible armored conduit. All internal wiring shall run to terminal boxes mounted on the condensing unit.

3.5.16 Automatic controls. Operation of chilled water units shall be completely automatic except as specified herein. The thermostat control shall start and stop the condensing unit motor and actuate the solenoid valve in the liquid refrigerant line. The high-low pressure cutout switch on the condensing unit shall be set as an additional safety device to prevent freeze-up and damage. A freeze-up control shall be provide which will sense the water temperature and upon reaching a temperature of 33°F, will stop the condensing unit, to prevent freeze-up of the tank.

3.5.16.1 Ingredient water controls. Controls for each ingredient water cooler shall include a remote bulb thermostat, with the bulb located in the storage tank. The electrically operated flow-control valves shall be so arranged that the water inlet will be shut off while chilled water from the cooler is being pumped to the dough mixer. At other times, the valves shall permit the recirculation of water within the cooler. A flow-control diagram shall be furnished with each cooler. All internal wiring of the flow-control valves shall be complete. Wiring diagrams shall be furnished.

3.5.16.2 Jacket water controls. The circulation of chilled water from the jacket water coolers shall be controlled by manual valves. The control for the jacket water cooler shall include a remote bulb thermostat, with the bulb located in the storage tank, and a solenoid valve in the liquid refrigerant line. The thermostat shall open and close the circuit to the solenoid when the temperature of the water in the cooler rises above or falls below a predetermined point. The thermostat and solenoid valve shall be so interconnected with the water cooler pump that the solenoid valve will not open unless the pump is operating.

3.5.16.3 Control box and instruments. A rigidly mounted, easily accessible control box, constructed of sheet steel and with a hinged door, shall be provided. Motor starting switches, relays, capacitors, high and low pressure controls, thermometers, and refrigerant pressure gages shall be installed in the



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control box. Each instrument shall be clearly identified by either a name plate or stenciling.

3.5.17 Thermometer. A dial thermometer conforming to MS 27211 shall be furnished with each water cooler to indicate temperatures of water in coolers, and shall be located so as to be easily read from the front of the unit.

3.6 Safety. All rotating or reciprocating parts and parts designed to operate at a temperature above 148°F, shall be insulated, fully inclosed, guarded, or otherwise isolated so that contact by operating personnel is not possible.

* 3.7 Electromagnetic compatibility. When specified (see 6.2), the chilled water units shall be designed and equipped for electromagnetic compatibility in accordance with class B, part 7 of MIL-STD-461 (see 4.4.4).

3.8 Finish. The chilled water units shall have the manufacturer's standard finish. The color used shall be as specified (see 6.2) and shall be uniform throughout.

* 3.9 Instructions. A chart containing printed instructions covering the operation and maintenance of the complete equipment shall be furnished. There shall also be furnished with each unit a printed schematic layout diagram for installation such as shown on Drawing No. 5-11-649. The chart and schematic shall be mounted in the control box (see 3.5.16.3).

* 3.10 Identification marking. The type, size and material for the identification plate shall conform to MIL-P-15024 when specified (see 6.2) and equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130, and shall include the following:

1. National Stock Number (NSN).
2. Manufacturer's nomenclature and identifying number.
3. Serial number with assigned suffix letter, when required by the contract.
4. Contract number (acquisition document identification number).
5. Government ownership designation (US).
6. Special characteristics, if applicable.
7. Manufacturer's code.

3.11 Workmanship. Rivets, when used, shall be driven to completely fill the holes with full sized heads neatly finished and in full contact with the surface of the member. Bolts, nuts, and screws shall be drawn tight and so secured that they shall not work loose under vibration.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document.

4.2.2 In-process inspection. Inspection shall be made of the following operations and requirements to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be to the items affected and to the operation.

a. Insulation installation conformance to requirements in 3.5.4 and 3.5.12.

b. Welding conformance to requirements in 3.5.2.

c. Closed pressure tank (when used) conformance to the leakage at 300 p.s.i.g. requirements in 3.5.5.

d. Water pump conformance to the applicable capacity requirement in 3.5.13.

* 4.2.3 End item visual examination. The end item shall be examined for the defects listed in table III. The lot size shall be expressed in units of chilled water units. The sample unit shall be one chilled water unit. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.



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TABLE III. End item visual defects

| Examine | Defect |
|---|---|
| Finish | Not finished, plated, or coated where required.<br>Type of finish not as specified; blistered, peeled, or chipped; or area of no film or thin film.<br>Color of finish not as specified or not uniform.<br>Sharp burr or splinters. |
| Water-cooler | Water cooler section not panel covered.<br>Ingredient water cooler flow control valve not electrical remote control.<br>Jacket water-cooler flow control valves not hand control.<br>No drain provided on tank.<br>Gravity drainage not complete.<br>Cooler not insulated.<br>Cooler not housed in insulated galvanized container.<br>Water cooler container not provided with access cover.<br>Thermometer not provided for water cooler. |
| Evaporator | Dehydrator does not have by-pass valve.<br>Liquid indicator not accessible.<br>Solenoid valve in liquid line has no manual provision.<br>No pressure gage at back pressure valve.<br>No by-pass provided back pressure valve. |
| Condensing unit | Condenser unit not vibration dampened.<br>Condenser unit not provided galvanized drip pan with lip and drain.<br>Condenser not provided drain.<br>Condenser end plates not removable.<br>Purging connection not provided. |
| Plumbing | Receiver inlet valve and outlet valve not provided.<br>Receiver valves not properly tagged.<br>Regulating valve not provided for closed jacket water cooler.<br>Drain not provided for water cooler.<br>Drain not extended below framework.<br>Chilled water lines not insulated.<br>Vibration or expansion loop not provided as required. |
| Pumps | Pump not vibration dampened.<br>Drain pan not provided drain cock. |



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TABLE III. End item visual defects (cont'd)

| Examine | Defect |
|---|---|
| | Pump not centrifugal type.<br>Pump not directly connected to motor.<br>Type II water pumps dissimilar. |
| Electrical | Motor controls not totally enclosed.<br>Motor starter not provided with overload relay.<br>Wiring not in conduit.<br>Internal terminal boxes not provided.<br>Manual valves not provided for chilled water circulation.<br>Control box not easily accessible.<br>Control not installed in control box. |
| Construction and workmanship, general (applicable to all components) | Fracture, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, or malformed.<br>Adjustable assembly that cannot be adjusted.<br>Component not readily accessible for servicing where required. |
| Welding | Scale, blisters, flux residue, or oxidation. |
| Riveting | Rivets loose or not completely driven; rivet heads not neatly finished or not in full contact with surface of member. |
| Bolts, nuts, and screws | Not adequately secured or tightened. |
| Marking identification and instructions | Not in accordance with MIL-STD-130. |

4.2.4 End item dimensional examination. The end item shall be examined for compliance with dimensions specified. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of chilled water units. The sample unit shall be one complete unit. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

* 4.2.5 End item testing.



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4.2.5.1 Representative item testing. One representative end item from the lot shall be tested as specified in 4.4.1 through 4.4.5. Any test failure shall be cause for rejection of the lot.

4.2.5.2 Leakage testing. Every end item shall be tested with a halide leak detector or other method approved by the contracting officer for conformance to 3.5.7. Any evidence of leakage shall be cause for rejection of the end item.

4.2.6 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE IV. Packaging inspection

| Examination | Defect |
|---|---|
| Markings, exterior and interior | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application. |
| Preservative | Preservative improperly applied or missing. |
| Materials | Components missing, damaged, or otherwise defective. |

* 4.3 Certification compliance examination. Certifications, certified test reports, or listing marks for standards that are submitted as proof of compliance with document requirements, shall be examined and validated (see 4.3.1).

* 4.3.1 ASHRAE standard. Prior to approval of the first lot, the contractor shall submit to the contracting officer or his authorized representative, satisfactory evidence that the chilled water units he proposes to furnish meets the applicable requirements of ASHRAE Standard No. 15. Acceptable evidence shall be a certified test report from a recognized independent test laboratory, accpetable to the Government indicating that the unit offered has been tested and conforms to the ASHRAE Standard No. 15.

4.4 Methods of inspection.

4.4.1 Operational and capacity test. The chilled water unit shall be tested for conformance with the requirements of 3.3.1 employing the procedure outlined in ASHRAE Standard No. 30. During conduct of the test it shall be observed that



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the volume of the water coolers meet the requirements of 3.3.1 and that the tanks do not leak water as specified in 3.5.5. Operation of condensing unit and circulating pump shall conform to 3.5.14.

4.4.2 Refrigeration moisture test. The refrigeration unit shall be operated in an ambient temperature of 90°F ± 2° or more for at least 1 hour. The moisture indicator shall then be observed to determine compliance with 3.5.7 as shown by the component manufacturer's color chart data.

4.4.3 Insulation tests.

* 4.4.3.1 Density test for insulation. When pre-expanded insulation is used, a 6-inch cube shall be cut and weighed to determine conformance to 3.4.4.2. When foamed-in-place insulation is used, one free foam sample of the insulation specified in 3.4.4.3, a minimum size 8-inch cube, shall be taken prior to each day's production. a 6-inch cube shall be cut from the center of the sample and weighed. This weight, 1/8 of a cubic foot, shall be multiplied by 8 to determine conformance to the density requirement of 3.4.4.3. Any nonconformance shall constitute failure of this test.

* 4.4.4 Electromagnetic compatibility test. When electromagnetic compatibility is required, the test unit shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461. Any nonconformance with the requirements specified shall constitute failure of this test.

4.4.5 Receiver test. When a condenser is furnished alone, the separate receiver shall be tested for conformance to the capacity requirements in 3.5.10.

5. Packaging.

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2.).

5.1.1 Level A.

5.1.1.1 Preservative. Cleaning, application and type of preservative, and method of preservation shall be in accordance with MIL-P-116 as specified herein.

5.1.1.1.1 Cleaning. All exposed, uncoated, ferrous metal surfaces of the unit shall be cleaned in accordance with process C-1 and thoroughly dried.



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5.1.1.1.2 Application of preservative. All cleaned, exposed, uncoated noncontacting ferrous metal surfaces (surfaces that do not contact other surfaces in operation) of the unit shall be coated with type P-1 preservative. All cleaned, exposed, uncoated contacting ferrous metal surfaces (surfaces that contact other surfaces in operation, except shafts and pulleys) of the unit shall be coated with type P-6 preservative. Shafts and pulleys shall be coated as specified in 5.1.1.1.4.

5.1.1.1.3 Protection of preservative. Surfaces from which preservative compound might be removed by contact with packaging and packing materials shall be covered or wrapped with greaseproof barrier material in accordance with method I.

5.1.1.1.4 Pulleys. Drive belts shall be removed, and exposed, uncoated, metal surfaces of the pulleys and shafts shall be coated with primer conforming to TT-P-664. When primer is thoroughly dry, drive belts shall be remounted in place with tension on the belts relaxed.

5.1.1.1.5 Lubrication. All bearings shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

5.1.1.1.6 Refrigeration system. The refrigerant charge on open type systems shall be pumped down, and the appropriate valves closed and capped. Closed valves shall be tagged or otherwise clearly marked with instructions for restoring to service. Warning tags shall be prominently affixed to the controls and shall caution against operating the unit without opening valves. Oil and refrigerant shall not be removed from the system. All floating mechanisms and tubing not rigidly fixed shall be secured in a fixed position to minimize vibration and subsequent damage during handling and shipment. Floating mechanism is defined as any part that is inherently flexible or movable, or any part that is rubber, spring, bracket, or otherwise mounted which might vibrate during shipment.

5.1.1.1.7 Protection of wiring. All exposed wiring shall be coated with varnish conforming to MIL-V-13811 unless fungicidal treatment has been applied. Care shall be exercised to see that the varnish is not applied to any electrical contact points or connections.

5.1.1.1.8 Sealing of openings. All openings into the motor switches, control boxes and pipes, and all name plates, identification plates, instruction plates, faces of gages, thermometer, or thermostats shall be sealed with tape conforming to type I or II, class 1 or PPP-T-60.

* 5.1.2 Commercial. Each chilled water unit shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).



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5.2.1 Level A packing. Each chilled water unit, preserved as specified in 5.1, shall be packed in a shipping container conforming to type I, class 1 or 2, style 2 of MIL-C-104. Cushioning, blocking and bracing, and anchoring or contents shall be in accordance with MIL-STD-1186.

5.2.2 Level B packing. Each chilled water unit, preserved and packaged as specified in 5.1, shall be packed in a shipping container conforming to MIL-C-3774. Cushioning, blocking and bracing, and anchoring of contents shall be in accordance with MIL-STD-1186.

5.2.3 Commercial packing. Chilled water units, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

* 6.1 Intended use. The type I and II chilled water units are intended for use in bakeries to provide chilled water for mixing ingredients. The type II has an additional intended use of providing chilled water or coolant to cooling jackets on mixing machines, covered by MIL-M-10921. The coolant shall be a mixture of water and propylene glycol, USP, to acquire a lower temperature.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Type and size required (see 1.2).
(c) Electrical characteristics (see 3.5.1)
(d) Type of condenser (see 3.5.7).
(e) When electromagnetic compatibility is required (see 3.7).
(f) Color of finish (see 3.8).
(g) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

* 6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.4).

6.4 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies to these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.



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DLA - GS

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Preparing activity:

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-10933E | Chilled Water Unit, Bakery |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

5/5

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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